University of Botswana

Department of Mathematics

Tutorial 4: Differentiation

Part I: Product Rule

In Exercises 1–5, use the Product Rule where applicable to find the first derivative or $\frac{dy}{dx}$.

1. $y = \frac{xe^x}{4}$
2. $y = \sqrt[5]{x^3} \sec x$
3. $f(x) = \frac{5}{8} e^{2x} \ln(\sin 3x)$
4. $y = (\tan x + \ln x + e^x)^2$
5. $y = (x^{\frac{2}{5}} - e^{-x}) \cot x$

Part II: Quotient Rule

In Exercises 6–10, use the Quotient Rule where applicable to find the first derivative or $\frac{dy}{dx}$.

6. $y = \frac{\sec x}{\tan x + 1}$
7. $g(x) = \frac{\ln \cos x}{e^{4x}}$
8. $f(t) = \frac{9 \sin t}{t+1}$
9. $y = \frac{e^{x^2}}{3 - 5x^3}$
10. Given that $f(x) = \frac{\cos x}{1 - \sin x}$ show that $\frac{df}{dx} = \frac{1}{1 - \sin x}$.

Part III: Higher Order Derivatives

For Question 11–13, find the given higher order derivative.

11. $f(x) = 2x - e^{-x}, \quad f'''(x)$
12. $f'(x) = \frac{3}{\sqrt[3]{x^2}} - 8, \quad f^{(4)}(x)$
13. $f^{(4)}(x) = \frac{\ln 3x}{5} - 3 \tan 2x, \quad f^{(5)}(x)$

Part IV: Chain Rule

In Exercises 14–20, find the derivative by using chain rule find the first derivative

14. $g(x) = (2 - a^2x + 3x^2)^{10}$
15. $f(x) = 7\sqrt[4]{(x - 5\sec 3x)^3}$
16. $f(t) = \frac{8}{\sqrt{9-4t^2}}$
17. $f(x) = \tan(\ln e^x)$
18. $f(x) = -5\cot(x + \sin x)$
19. $f(x) = 3\cos^4 7x$
20. Given that $y = u^3$ and $u = 8x^5 - 4x^3$ find $\frac{dy}{dx}$